

PROJECT DESIGN PHASE

SOLUTION ARCHITECTURE

Date	29 September 2026
Team ID	6ab800af30f9767f87729b97
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	4 Marks

Solution Architecture:

The solution architecture bridges the project access-control problem and the ServiceNow technology solution. It describes the structure and behaviour of the ACL-controlled application, the access decision flow, and the record operations protected by the solution.

Architecture Objective: Provide controlled access to the Institution Details table by evaluating the user's role and the configured record/Branch condition before allowing the requested operation.

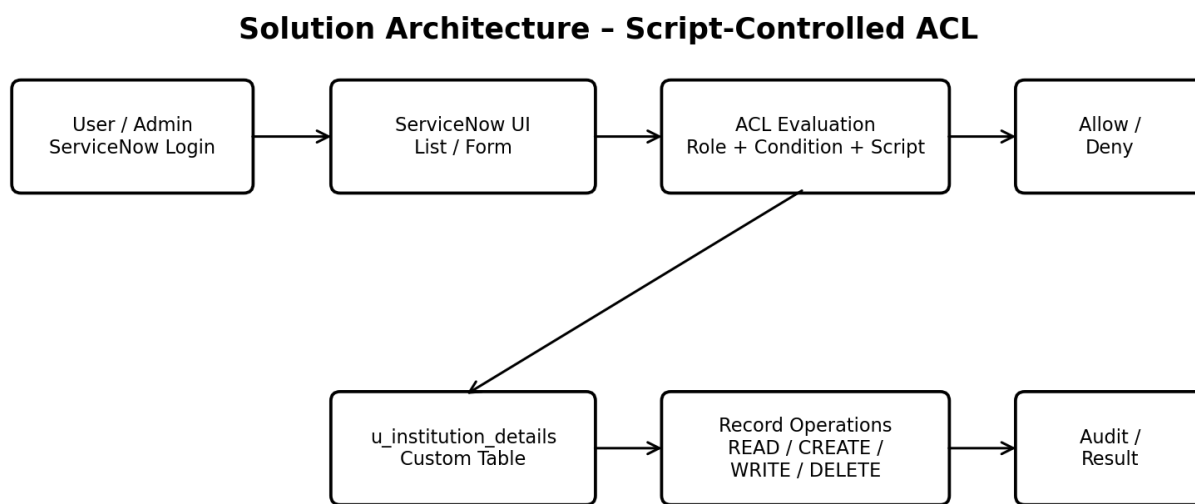
User and Interface Layer: Users and administrators access the ServiceNow application through the list and form interfaces for the Institution Details table.

Security / ACL Layer: ServiceNow evaluates the applicable ACL. The project uses a script-controlled READ ACL and separate CREATE, WRITE and DELETE ACLs. The configured roles are bb1, bb2, bb3 and bb4.

Data Layer: The custom table u_institution_details stores Institution Details records including Student Roll Number, Student Name, Faculty Name, Branch, Email, Phone Number and Description.

Authorization Decision: The ACL evaluation produces an allow or deny result. Administrators are configured for full access, while the intended EEE-user scenario is restricted according to the project ACL rules.

Figure 1: Solution Architecture and Access-Control Flow



Access request → ServiceNow ACL evaluation → authorization decision → permitted operation/result

Figure explanation: A user or administrator sends an access request through the ServiceNow UI. The ACL layer evaluates the configured role, condition and script. The request is then allowed or denied, and permitted operations are applied to the u_institution_details table.

Access-Control Mapping

Operation	ACL Role	Purpose	Expected Access
READ	bb1 + configured READ condition/script	Control viewing of records	Permitted EEE-user scenario / Admin
CREATE	bb2	Control new-record creation	Authorized users
WRITE	bb3	Control record modification	Authorized users
DELETE	bb4	Control record deletion	Authorized users

Architecture Note: The supplied template requires a solution architecture and provides an example architecture diagram. The diagram above is project-specific and represents the actual ServiceNow ACL flow; the unrelated example diagram from the template has not been copied.